Solutions, Twelfth Annual ECC Undergraduate Mathematics Competition, April 4, 2009

1. Solve for x.

The solutions are $\boxed{x = \pm\sqrt{2}/4}$. The following are equivalent:

$$\log(4^3x^25^3)^{-3} = -9;$$
$$-3\log(4^3x^25^3) = -9;$$
$$\log(4^3x^25^3) = 3;$$
$$4^3x^25^3 = 10^3;$$
$$x^2 = \frac{10^3}{20^3} = \frac{1}{8}.$$

Thus

$$x = \pm\sqrt{\frac{1}{8}} = \pm\frac{\sqrt{2}}{4}.$$

2. A multiple of 5.

Since $4 \equiv -1 \pmod 5$ and $3 \equiv -2 \pmod 5$, we have for every odd integer $2k-1$ that

$$4^{2k-1} \equiv -1 \pmod 5 \quad \text{and} \quad 3^{2k-1} \equiv (-2)^{2k-1} = -2^{2k-1},$$

so $2^{2k-1} + 3^{2k-1} \equiv 0 \pmod 5$. Thus

$$1^{2k-1} + 2^{2k-1} + 3^{2k-1} + 4^{2k-1} + 5^{2k-1} \equiv 0 \pmod 5.$$

3. Minimum value.

The minimum value is $\boxed{1/180}$, occurring at $y = -1/6$. We have

$$\begin{aligned} f'(y) &= -2\int_0^1 (x^2 - x - y)dx \\ &= -2\left(\frac{x^3}{3} - \frac{x^2}{2} - xy\right)\Big|_0^1 \\ &= -2\left(\frac{1}{3} - \frac{1}{2} - y\right) \\ &= 2y + \frac{1}{3}. \end{aligned}$$

Thus, $f'(y) = 0$ when $y = -1/6$, $f'(y)$ is negative for $y < -1/6$ and positive for $y > -1/6$, so $f(y)$ has a global minimum at $y = -1/6$. The minimum value is

$$f\left(\frac{-1}{6}\right) = \int_0^1 \left(x^2 - x + \frac{1}{6}\right)^2 dx = \frac{1}{180}.$$

4. Tower of Babel.

It would be $\boxed{184{,}320 \text{ meters}}$ high. Let $a = 1 + 2 + \cdots + 9 = 45$. In years 1 through 9 the tower grows a meters. In years 11 through 99 it grows $a(1 + 2 + \cdots + 9) = a^2$ meters. In years 111 through 999 it grows $a^2(1+2+\cdots+9) = a^3$ meters. In years 1111 through 1999 it again grows a^3 meters, and in all other years the digit 0 occurs and there is 0 growth. Thus the height at the end of year 2009 is $45 + 45^2 + 2 \cdot 45^3 = 184,320$ meters.

5. Sum of squares of the roots.

We show that $\boxed{r^2 + s^2 + t^2 = 19}$. Equating coefficients of x and of x^2 in the equation

$$(x - r)(x - s)(x - t) = x^3 + 3x^2 - 5x - 1$$

we have

$$r + s + t = -3 \quad \text{and} \quad rs + rt + st = -5,$$

so

$$9 = (r + s + t)^2 = r^2 + s^2 + t^2 + 2(rs + rt + st) = r^2 + s^2 + t^2 + 2(-5).$$

Thus $r^2 + s^2 + t^2 = 9 + 10 = 19$.

6. Urns and balls.

The probability is $\boxed{13/14}$. There are $\binom{8}{4} = 70$ ways to choose four balls from eight. In how many of these are there fewer than three red balls? There are $\binom{3}{2}\binom{5}{2} = (3)(10)$ ways to get exactly two red and two white balls, $\binom{3}{1}\binom{5}{3} = (3)(10)$ ways to get one red and three white, and $\binom{5}{4} = 5$ ways to get no reds and four whites. Thus the probability that fewer than three red balls are obtained in four draws is

$$\frac{(3)(10) + (3)(10) + 5}{70} = \frac{65}{70} = \frac{13}{14}.$$

ALTERNATE SOLUTION

The probability that all 3 reds are included in the first four draws is

$$\frac{\binom{3}{3}\binom{5}{1}}{\binom{8}{4}} = \frac{5}{70} = \frac{1}{14},$$

so the probability that it takes more than four draws is $\frac{13}{14}$.

7. A double sequence.

The value is $\boxed{-1}$. We show by induction that for all n, $x_n^2 - 2y_n^2 = (-1)^n$. We have $x_0^2 - 2y_0^2 = 1 = (-1)^0$. Suppose that $x_k^2 - 2y_k^2 = (-1)^k$. Then

$$\begin{aligned} x_{k+1}^2 - 2y_{k+1}^2 &= (x_k + 2y_k)^2 - 2(x_k + y_k)^2 = x_k^2 + 4x_k y_k + 4y_k^2 - 2(x_k^2 + 2x_k y_k + y_k^2) \\ &= -x_k^2 + 2y_k^2 = (-1)(x_k^2 - 2y_k^2) = (-1)(-1)^k = (-1)^{k+1}. \end{aligned}$$

By induction, the claim follows. In particular, $x_{2009}^2 - 2y_{2009}^2 = (-1)^{2009} = -1$.

8. Area of a triangle.

The area is $\boxed{960/\sqrt{7}}$. Let the sides of the triangle be a, b, c, with 20 being the altitude to side a, 24 to side b and 30 to side c. Then the area of the triangle is given by $A = 10a = 12b = 15c$, so $b = 5a/6$, and $c = 2a/3$. The area is also given by

$$A = \sqrt{s(s-a)(s-b)(s-c)},$$

where

$$s = \frac{a+b+c}{2} = \frac{1}{2}\left(a + \frac{5a}{6} + \frac{2a}{3}\right) = \frac{5a}{4}.$$

This gives us

$$A = \sqrt{\left(\frac{5a}{4}\right)\left(\frac{5a}{4} - a\right)\left(\frac{5a}{4} - \frac{5a}{6}\right)\left(\frac{5a}{4} - \frac{2a}{3}\right)} = \sqrt{\left(\frac{5}{4}\right)\left(\frac{1}{4}\right)\left(\frac{5}{12}\right)\left(\frac{7}{12}\right)a^4} = \frac{5\sqrt{7}a^2}{48}.$$

Thus $10a = A = 5\sqrt{7}a^2/48$, so $a = 96/\sqrt{7}$ and the area is $A = 10a = 960/\sqrt{7}$.

9. Convergence of a series.

Let $z_k = x_k + iy_k$ with x_k and y_k real. Then the convergence of

$$\sum z_k = \sum x_k + i \sum y_k$$

implies that $\sum x_k$ and $\sum y_k$ both converge. Likewise, the convergence of

$$\sum z_k^2 = \sum (x_k^2 - y_k^2) + 2i \sum x_k y_k$$

implies that $\sum (x_k^2 - y_k^2)$ and $\sum x_k y_k$ both converge. Because $x_k \geq 0$, the convergence of $\sum x_k$ implies the convergence of $\sum x_k^2$, by the comparison test. Thus

$$\sum |z_k|^2 = \sum (x_k^2 + y_k^2) = 2 \sum x_k^2 - \sum (x_k^2 - y_k^2)$$

converges, Q.E.D.

10. Solutions in integers.

The only solutions are $\boxed{(m,n) = (3,5) \text{ and } (m,n) = (4,7)}$. These are easily found by trial; it remains to show there are no others. Write the equation in the equivalent form

$$3 \cdot 2^m = (n-1)(n+1).$$

From this we see that one of $n+1$ and $n-1$ must be a multiple of 3. More specifically, either

(i) $n+1 = 3 \cdot 2^a$ and $n-1 = 2^b$, or

(ii) $n-1 = 3 \cdot 2^a$ and $n+1 = 2^b$

for some nonnegative integers a and b with $a+b = m > 0$. In case (i),

$$3 \cdot 2^a = n+1 = (n-1)+2 = 2^b + 2.$$

If $a = 0$ then $b = 0$ also, and $m = a+b$ would be 0. Thus, for $m > 0$ we need $a > 0$. Then

$$3 \cdot 2^{a-1} = 2^{b-1} + 1.$$

For $2^{b-1}+1$ to be a multiple of 3 we must have $b \geq 2$. Then $2^{b-1}+1$ is odd, so 2^{a-1} must be odd. Thus $a = 1$, making $b = 2, n = 5$, and $m = 3$.

In case (ii) we have

$$2^b = n+1 = (n-1)+2 = 3 \cdot 2^a + 2,$$

so $b > a$. Thus $b \geq 1$ and 2^b is even. It follows that $3 \cdot 2^a$ is even, so $a \geq 1$, and we have

$$2^{b-1} = 3 \cdot 2^{a-1} + 1.$$

From this we see that $2^{b-1} > 1$, whence $b \geq 2$ and 2^{b-1} is even. It follows that $3 \cdot 2^{a-1}$ is odd, so $a = 1$. Thus $b = 3, m = 4$ and $n = 7$. This shows that (3,5) and (4,7) are the only solutions in positive integers.